

Sense of Agency on Handheld AR for Virtual Object Translation

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ABSTRACT

Handheld augmented reality (AR) interfaces are applied in various programs nowadays, and the degrees of freedom (DoF) of translation modes have become significant on these interfaces. Sense of agency (SoA), emphasizing one's feeling of control, has emerged as an essential index of user experience. However, little was known about users' feelings of control with different translation modes in literature. Hence, this paper focuses on users' SoA in different translation modes by assessing subjective and objective measures. Correlations between SoA and translation modes were explored on handheld AR interfaces, revealing that the 1DoF translation mode was associated with higher SoA.

Keywords: Handheld augmented reality, virtual object translation, sense of agency, electroencephalogram.

Index Terms: H.1.2 [Human-centered computing]: Human computer interaction (HCI) — HCI design and evaluation methods — User studies.

1 INTRODUCTION

Rapid innovations in augmented reality (AR) technology enable interactions between the real and virtual worlds on handheld mobile devices. Handheld AR, which refers to AR systems based on mobile devices, is well suited for implementation with touch-based interactive approaches [1].

Virtual object translation is a critical feature on handheld AR interfaces. Scholars investigated the effects of two translation modes, degrees of freedom (DoF) separation and DoF integration, on users' task performance and psychological experience in handheld AR settings [1, 2]. DoF separation refers to the action of restricting virtual objects to translate on the x-, y-, or z-axes, whereas DoF integration is the operation of synchronically translating virtual objects on all axes [3].

As an essential index of user experience in the field of human-computer interaction (HCI), sense of agency (SoA) emphasizes one's feelings of control of own actions [4]. Previous literature indicated that human behavior and mental states rely heavily on SoA, which is crucial for motor control and action selection [5]. Specifically, users prefer actions associated with higher SoA over those that are not [4]. Previous research evaluated SoA mainly using self-reports along with task performance [5]. However, little is known about the influence of virtual object translation with a series of actions on SoA, especially when measured with electroencephalogram (EEG) data [4].

Therefore, this study explores how virtual object translation affects users' SoA on handheld AR interfaces through self-report, task performance, and EEG data. The contributions of this study are listed as follows: (i) investigating the relationship between virtual object translation modes and SoA with subjective and

objective measures; (ii) exploring the translation mode with higher SoA on handheld AR interfaces by concentrating on users' feelings of control.

2 USER STUDY

Thirty-two participants (16 males and 16 females) from 20 to 33 years ($M = 23.4$, $SD = 5.43$) were recruited for this study. They were informed of general information about this study, and all of them signed the consent form before the experiment. This study has received ethical approval from the University Ethics Committee.

The experimental setup, as displayed in Figure 1 (i), consisted of (a) a 32-channel EEG head cap to record the EEG data and (b) a 12.7-inch touch-based tablet with AR application installed.

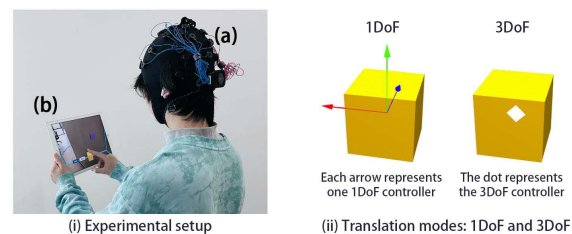


Figure 1: Experimental design: (i) experimental setup including (a) EEG cap and (b) touch-based tablet; and (ii) two virtual object translation modes.

During the experiment, participants were asked to apply 1DoF and 3DoF modes depicted in Figure 1 (ii) to complete translation tasks in the simulated environment on handheld AR interfaces.

- 1DoF translation mode. Participants used the one-finger gesture to select one controller (one of the three arrows) and moved the finger across the tablet screen to translate the object on the x-, y-, or z-axes respectively.
- 3DoF translation mode. Participants used the one-finger gesture to select the controller (one dot) and moved the finger to translate the object on the tablet screen on all axes synchronically.

Before the experiment, participants had the chance to adapt to 1DoF and 3DoF translation modes. Then, they were asked to complete three tasks for each translation mode. In each task, one movable object and one target position were displayed in the virtual environment. Participants needed to translate the movable object as precisely as possible to the target position without time restriction. During the experiment, participants could walk around in a 5m×3.5m physical space while holding the tablet and interacting with the simulated world on the touch-based tablet.

This experiment aimed to examine users' SoA while conducting translation tasks in handheld AR applications by assessing several measures listed below.

- Subjective measures. A widely-used questionnaire with four items [4] was adopted to obtain participants' subjective SoA.
- Performance measures. The manipulation time and error rate of translation tasks, both of which can reflect SoA from a different perspective, were recorded by Unity program automatically.

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- EEG measures. Spectral power analysis and brain activity analysis were applied to explore SoA based on EEG data recorded by the EEG cap from subjects.

3 RESULTS

Data from self-reports revealed a downward trend in average SoA score from 1DoF to 3DoF mode on handheld AR interfaces without significant differences after applying the Kruskal-Wallis test. Additionally, based on the analysis of performance data using the Kruskal-Wallis test, participants spent an average of twenty seconds longer using 3DoF mode than 1DoF mode, whereas there was no statistically significant difference in error rate analysis.

EEG spectral power analysis was used to compute alpha, beta, and gamma power, which were connected to SoA evaluation. The Wilcoxon signed-rank test revealed significant increases in alpha power throughout the whole brain from 1DoF to 3DoF modes, as shown in Figure 2. Significant differences in beta power were observed in certain regions when participants employed 1DoF mode compared with 3DoF mode. Besides, significant increases in gamma power were mainly observed from 1DoF to 3DoF across the whole brain.

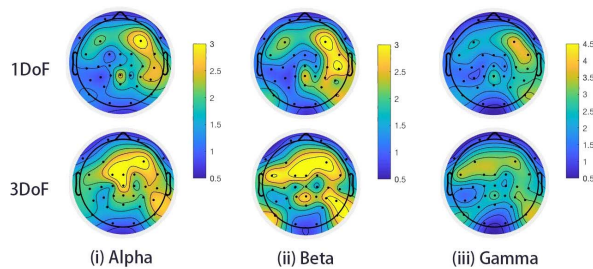


Figure 2: Spectral power analysis: the brain topographic map of (i) alpha, (ii) beta, and (iii) gamma band power.

Alpha, beta, and gamma phase coherences were calculated to represent brain connectivity associated with SoA. Figure 3 shows that alpha coherences increased significantly across different regions from 1DoF to 3DoF mode. The inverse trend was found in the beta band between specific regions. Besides, although the Wilcoxon signed-rank test indicated that the gamma coherences increased dramatically from 1DoF to 3DoF in some regions, they remained unstable.

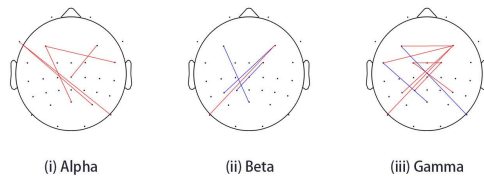


Figure 3: Brain connectivity between 1DoF and 3DoF modes: the brain connectivity map of (i) alpha, (ii) beta, and (iii) gamma phase coherences (red line indicates significant phase coherence increases and blue line indicates phase coherence decreases from 1DoF to 3DoF modes).

4 DISCUSSION

Self-reports could directly represent participants' subjective feelings of control in simulated environments [4]. This study explicitly revealed that 1DoF mode was related to higher SoA

scores, resulting in better feelings of control on handheld AR interfaces.

Based on performance measures, participants in this study spent less manipulation time completing translation tasks in 1DoF mode than in 3DoF mode. Previous research presented that people performed more effectively with increased SoA [4]. This study revealed that participants performed effectively in 1DoF mode, possibly due to enhanced SoA in 1DoF mode.

EEG measures have been proven to be effective in assessing SoA. Prior literature found a negative relationship between frontal alpha power and changes in SoA [5]. Besides, frontal lobes are related to brain networks that enable motor control [6]. This study found a downward trend in frontal alpha power from 1DoF to 3DoF mode, possibly due to reduced frontal lobe activation in 1DoF mode. Furthermore, previous literature indicated that alpha coherences should be positively connected with information processing in the brain [5]. This paper revealed that 1DoF mode with less brain activation resulted in higher SoA on handheld AR interfaces, as participants generated fewer alpha coherences during applying 1DoF mode.

5 CONCLUSION

This study focused on SoA evaluation by concentrating on users' feelings of control in 1DoF and 3DoF translation modes on handheld AR interfaces. The results presented that 1DoF mode contributed to higher SoA compared with 3DoF mode based on self-reports, task performance, and EEG data from subjects. 1DoF mode could be a better option for translation tasks on handheld AR interfaces, when greater feelings of control are preferred.

Besides virtual object translation, other manipulation types, such as rotation and scaling, could be explored in further studies. These manipulation types are also essential on handheld AR interfaces, and users' feelings of control is significant for understanding the user experience during the manipulation.

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